4 Monitoring and Operating the Controller

4.1 Overview

Introduction

This section gives you all the information necessary to help you monitor and operate your controller including an Operator Interface overview, how to lockout changes to the controller, entering a security code, and monitoring the displays.

What's in this section?

The following topics are covered in this section.

4.2 Operator Interface

Introduction

Figure 4-1 is a view of the Operator Interface.

Figure 4-1 Operator Interface

4.3 Entering a Security Code

Introduction

The level of keyboard lockout may be changed in the Set Up mode. However, knowledge of a security code number (0 to 9999) may be required to change from one level of lockout to another. When a controller leaves the factory, it has a security code of 0 which permits changing from one lockout level to another without entering any other code number.

Procedure

If you require the use of a security code, select a number from 0001 to 9999 and enter it when the lockout level is configured as NONE. Thereafter, that selected number must be used to change the lockout level from something other than NONE.

ATTENTION Write the number on the Configuration Record Sheet in the configuration section so you will have a permanent record.

Table 4-1 Procedure to Enter a Security Code

Step	Operation	Press	Result
1	Enter Set Up Mode	Setup	*Upper Display* = **SET UP** *Lower Display* = **TUNING**
2	Select any Set Up Group	Function	*Upper Display* = **0** *Lower Display* = **SECUR**
3	Security Code Entry	▲ or ▼	To enter a four digit number in the upper display (0001 to 9999) This will be your security code.

4.4 Lockout Feature

Introduction

The lockout feature in the UDC3200 is used to inhibit changes (via keyboard) of certain functions or parameters by unauthorized personnel.

Lockout levels

There are different levels of Lockout depending on the level of security required. These levels are:

- **NONE** No Lockout. All groups Read/Write.
- **CALIB** Calibration prompts are deleted from the Setup List.
- **+CONFIG** Timer, Tuning, SP Ramp, and Accutune are Read/Write. All other Setup are Read only. Calibration Group is not available.
- **+VIEW** Timer, Tuning, and SP Ramp are Read/Write. No other parameters are available.
- **ALL** Timer, Tuning, and SP Ramp are Read only. No other parameters are viewable.

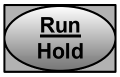
See *Subsection 3.4 - Tuning Parameters Set Up Group* prompts to select one of the above.

Security Code *(see Subsection 4.3)*

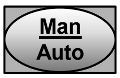
Individual key lockout

There are three keys that can be disabled to prevent unauthorized changes to the parameters associated with these keys. *First set the "Lock" prompt to NONE.*

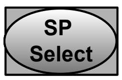
These keys are:



Key - you can disable the Run/Hold key for Set Point Programming at configuration Set Up group prompt "Tuning," function prompt "RN HLD."



Key - you can disable the Auto/Manual key at configuration Set Up, group prompt "Tuning", function prompt "AUTOMA"



Key - you can disable the Set Point Select function key at configuration Set Up group prompt "Tuning," function prompt "SP SEL."

See *Subsection 3.4 - Tuning Parameters Set Up Group* prompts to enable or disable these keys.

Key error

When a key is pressed and the prompt "Key Error" appears in the lower display, it will be for one of the following reasons:

- Parameter not available or locked out
- Not in setup mode, press **SET UP** key first
- Individual key locked out.

4.5 Monitoring Your Controller

4.5.1 Annunciators

The following annunciator functions have been provided to help monitor the controller:

Table 4-2 Annunciators

Annunciator	Indication
ALM 1 2	*A visual indication of each alarm* Blinking 1 indicates an alarm latched condition. The blinking will continue even after the alarm condition ends until it is acknowledged by pressing the RUN/HOLD key.
OUT 1 2	*A visual indication of the control relays*
DI 1 2	*A visual indication of each Digital Input*
A or **MAN**	*A visual indication of the mode of the controller)* **A**—Automatic Mode **MAN**—Manual Mode
[None], **F** or **C**	*A visual indication of the temperature units* **[None]**—No temperature unit annunciator **F**—Degrees Fahrenheit **C**—Degrees Celsius
■	*A visual Lamp to indicate when the lower display is showing the Active Setpoint (Local 1, Local 2, Local 3, Remote Setpoint or Computer Setpoint)*
	The upper left digit of the display is used to show other annunciator functions **T**—Accutuning in process **C**—Computer overide active **O**—Output override active

4.5.2 Viewing the operating parameters

Press the **LOWER DISPLAY** key to scroll through the operating parameters listed in Table 4-3. The lower display will show only those parameters and their values that apply to your specific model.

Table 4-3 Lower Display Key Parameter Prompts

Lower Display	Description
OUT XX.X	OUTPUT—Output value is shown in percent with one decimal point for all output types except Three Position Step Control (TPSC). For TPSC, when no slidewire is connected, this display is an estimated motor position and is shown with no decimal point. For Position Proportional Control, if the slidewire fails, then the instrument automatically switches over to TPSC and the OUT display changes with it.
SP XXXX	LOCAL SETPOINT #1—Also current setpoint when using SP Ramp.
2SP XXXX	LOCAL SETPOINT #2
3SP XXXX	LOCAL SETPOINT #3
RSP XXXX	REMOTE SETPOINT
1IN XXXX	INPUT 1—Used only with combinational input algorithms.
2IN XXXX	INPUT 2
POS XX	SLIDEWIRE POSITION—Used only with TPSC applications that use a slidewire input.
CSP XXXX	COMPUTER SETPOINT—When SP is in override.
DEV XXXX	DEVIATION—Maximum negative display is –999.9.
PIDSET X	TUNING PARAMETER —where X is either 1 or 2.
ET HR.MN	ELAPSED TIME—Time that has elapsed on the Timer in Hours.Minutes.
OTR HR.MN	TIME REMAINING—Time remaining on the Timer in Hours.Minutes. The “O” is a rotating clock face.
RAMPXXXM	SETPOINT RAMP TIME—Time remaining in the Setpoint Ramp in minutes.
SPN XXXX	SETPOINT NOW—Current Setpoint when SP Rate is enabled. The **SP XXXX** display shows the “target” or final setpoint value.
XXRAHR.MN	RAMP SEGMENT NUMBER AND TIME REMAINING—Set Point Programming display. XX is the current segment number and **HR.MN** is the time remaining for this segment in Hours.Minutes.
XXSKHR.MN	SOAK SEGMENT NUMBER AND TIME REMAINING— Set Point Programming display. XX is the current segment number and **HR.MN** is the time remaining for this segment in Hours.Minutes.
RECYC XX	NUMBER OF SP PROGRAM RECYCLES REMAINING
To BEGIN	RESET SP PROGRAM TO START OF FIRST SEGMENT
RERUN	RESET SP PROGRAM TO START OF CURRENT SEGMENT
AUX XXXX	AUXILIARY OUTPUT—Displayed only when output algorithm is not Current Duplex.
BIA XXXX	BIAS—Displays the manual reset value for algorithm PD+MR.
TUNE OFF	LIMIT CYCLE TUNING NOT RUNNING—Appears when Accutune is enabled but not operating.
DO FAST	Limit Cycle Tuning with the objective of producing quarter-damped tuning parameters. This tuning may result in PV overshoot of the SP setting.
DO SLOW	Limit Cycle Tuning with the objective of producing damped or Dahlin tuning parameters, depending upon the detected process deadtime. The tuning parameters calculated by this selection are aimed at reducing PV overshoot of the SP setting.

4.5.3 Diagnostic Messages

The UDC3200 performs background tests to verify data and memory integrity. If there is a malfunction, a diagnostic message will be shown on the lower display. In the case of more than one simultaneous malfunction, only the highest priority diagnostic message will be displayed. Table 4-4 shows the error messages in order by priority. If any of these diagnostic messages appear on the lower display, refer to *Section 7 - Troubleshooting* for information on how to correct the problem.

Table 4-4 Diagnostic Messages

Prompt	Description
EE FAIL	Unable to write to nonvolatile memory. A subsequent successful write to nonvolatile memory removes this message.
FAILSAFE	This error message appears whenever the controller goes into a failsafe mode of operation. Failsafe operation occurs when an analog input fails or when configuration is corrupted.
INP1FAIL	Two consecutive failures of input 1 integration or input value is outside of Out-of-Range limits.
INP2FAIL	Two consecutive failures of input 2 integration or input value is outside of Out-of-Range limits.
SW FAIL	Slidewire input failure. Position Proportional Control automatically switched to Three Position Step Control.
CONF ERR	Configuration Errors—Low limit greater than high limit for PV, SP, Reset, or Output.
SOOTING	Carbon Potential Problem—Percent Carbon outside of “sooting boundary.”
IN1 RNG	Input 1 Out-of-Range—Input is outside of the High or Low Limits. Out-of-range criteria: Linear range: ± 10 % Characterized range: ± 1 %
IN2 RNG	Input 2 Out-of-Range—Same criteria as Input 1.
PV LIMIT	PV beyond limits—PV outside of PV Hi/Low Limits. PV = (PV source x PV source ratio) + PV source bias
FAILSAFE	Failsafe—Check inputs or configuration.
RV LIMIT	Remote Variable beyond limits—RSP outside of SP Hi/Low Limits. RV = (RV source x RV source ratio) + RV source bias
SEG ERR	Segment Error—SP Program starting segment number is less than ending segment number.
CAL MTR	Slidewire not calibrated. Perform Slidewire calibration.
SW FAIL	Position Proportional slidewire input failure.
TC1 WARN	The Thermocouple on Input 1 is starting to burnout.
TC2 WARN	The Thermocouple on Input 2 is starting to burnout.
TC1 FAIL	Thermocouple on Input 1 is in imminent danger of burning out.

Prompt	Description
TC2 FAIL	Thermocouple on Input 2 is in imminent danger of burning out.
OUT1FAIL	Current Output is less than 3.5 mA.
OUT2FAIL	Current Output 2 is less than 3.5 mA.

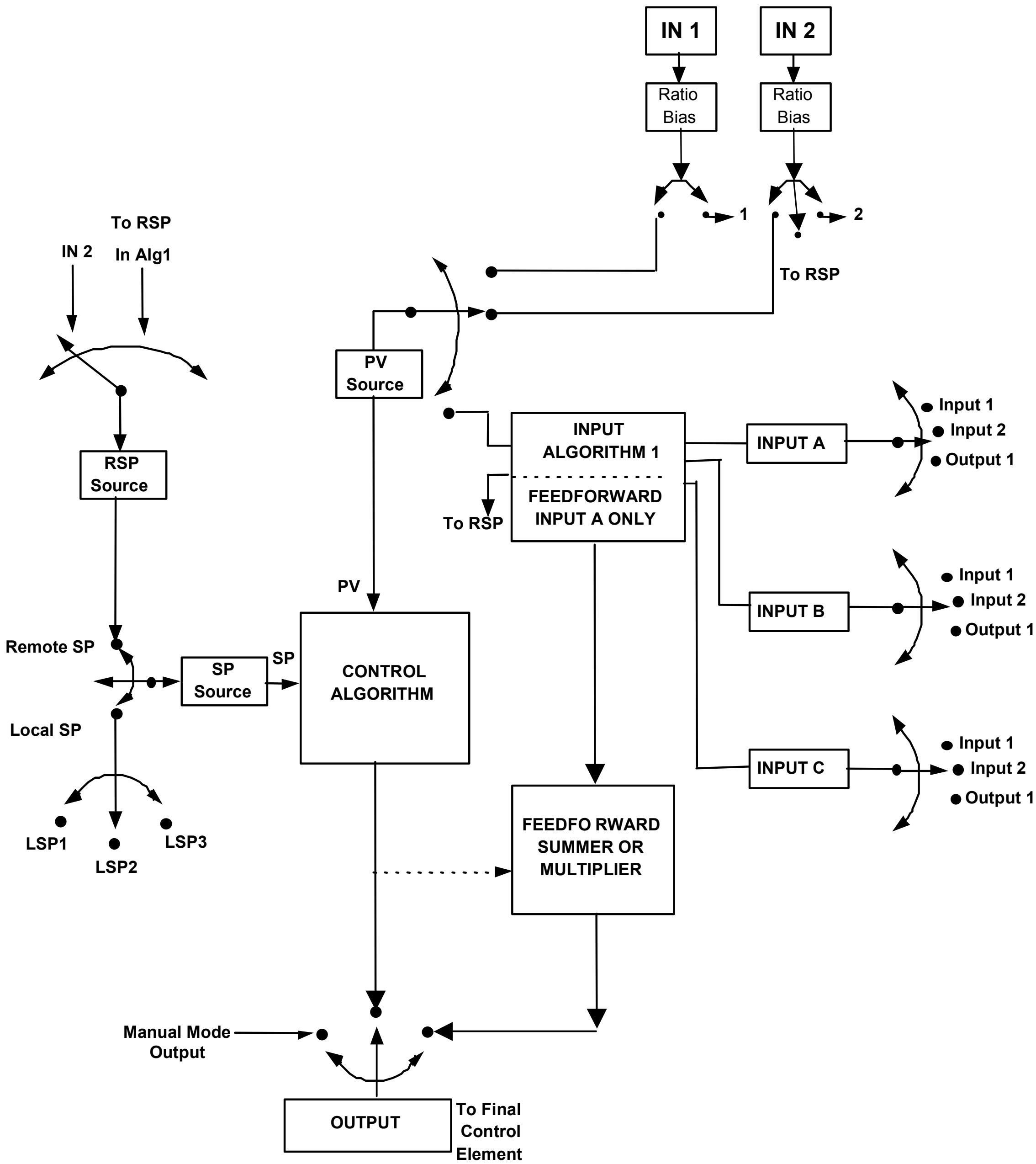


Figure 4-2 Functional Overview Block Diagram of the UDC3200 Controller

4.6 Start Up Procedure for Operation

Table 4-5 Procedure for Starting Up the Controller

Step	Operation	Press	Result
1	Select Manual Mode	**Man Auto**	Until "**M**" indicator is ON. The controller is in manual mode.
2	Adjust the Output	▲or▼	To adjust the output value and ensure that the final control element is functioning correctly. *Upper Display* = **PV Value** *Lower Display* = **OUT and the output value in %**
3	Enter the Local Setpoint	**Lower Display** ▲or▼	*Upper Display* = **PV Value** *Lower Display* = **SP and the Local Setpoint Value** To adjust the local setpoint to the value at which you want the process variable maintained. The local setpoint cannot be changed if the Setpoint Ramp function is running.
4	Select Automatic Mode	**Man Auto**	Until "**A**" indicator is ON. The controller is in Automatic mode. The controller will automatically adjust the output to maintain the process variable at setpoint.
5	Tune the Controller	**Setup**	Make sure the controller has been configured properly and all the values and selections have been recorded on the Configuration Record Sheet. Refer to Tuning Set Up group to ensure that the selections for PBor GAIN, RATE T, and I MIN, or I RPM have been entered. **Use Accutune to tune the controller; see the procedure in this section.**

4.7 Control Modes

ATTENTION

After changing a Local Setpoint value, if no other key is pressed, it then takes a minimum of thirty (30) seconds elapsed time before the new value is stored in non-volatile memory. If controller power is removed before this time, the new setpoint value is lost and the previous setpoint value is used at power-up. If, after changing the LSP value, another key is pressed, then the value is stored immediately.

4.7.1 Mode Definitions

Table 4-6 Control Mode Definitions

Control Mode	Definition
AUTOMATIC with LOCAL SETPOINT	In automatic local mode, the controller operates from the local setpoints and automatically adjusts the output to maintain the PV at the desired value. In this mode you can adjust the setpoint. See *Subsection 4.8 - Setpoints.*
AUTOMATIC with REMOTE SETPOINT (optional)	In automatic remote mode, the controller operates from the setpoint measured at the remote setpoint input. Adjustments are available to ratio this input and add a constant bias before it is applied to the control equation. See *Subsection 3.9 Input 1 or 3.10 Input 2.*
MANUAL (optional)	In the manual mode, the operator directly controls the controller output level. The process variable and the percent output are displayed. The configured High and Low Output Limits are disregarded and the operator can change the output value, using the increment and decrement keys, to the limits allowed by the output type (0 % to 100 % for a time proportioning output or –5 % to 105 % for a current output).

4.7.2 What happens when you change modes

Table 4-7 Changing Control Modes

Control Mode	Definition
Manual to Automatic Local Setpoint	The Local Setpoint is usually the value previously stored as the Local Setpoint. PV tracking is a configurable feature which modifies this. For this configuration, when the controller is in manual mode, the local setpoint value tracks the process variable value continuously.
Manual or Auto Local to Automatic Remote SP	The Remote Setpoint uses the stored ratio and bias to calculate the control setpoint. Auto bias is a configurable feature which modifies this. When it is selected the transfer from automatic local to automatic remote or from manual remote to automatic remote adjusts the bias based on the local setpoint such that Bias = LSP – (RSP Input x R).
Automatic Remote Setpoint to Manual or Auto Local Setpoint	If configured for local setpoint tracking, RSP, when the controller transfers out of remote setpoint the last value of the remote setpoint is inserted into the local setpoint. If LSP tracking is not configured, the local setpoint will not be altered when the transfer is made.

4.8 Setpoints

Introduction

You can configure the following setpoints for the UDC3200 controller.

- A Single Local Setpoint
- 2 Local Setpoints
- 3 Local Setpoints
- Up to 3 Local Setpoint and one Remote Setpoint

Refer to *Subsection 3.11 – Control Set Up Group* for configuration details.

Changing the Setpoints

Table 4-8 Procedure for Changing the Local Setpoints

Step	Operation	Press	Result
1	Select the Setpoint	Lower Display	Until you see: *Upper Display* = **PV** *Lower Display* = **SP or 2SP or 3SP (Value)**
2	Change the Value	▲ or ▼	To change the Local Setpoint to the value at which you want the process maintained. The display “blinks” if you attempt to enter setpoint values beyond the high and low limits..
3	Return to PV Display	Lower Display	To store immediately or will store after 30 seconds.

Switching between setpoints

You can switch Local and Remote setpoints or between two Local setpoints when configured.

ATTENTION The REMOTE SETPOINT value cannot be changed at the keyboard.

Table 4-9 Procedure for Switching Between Setpoints

Step	Operation	Press	Result
1	Select the Setpoint	SP Select	To switch between the Three Local Setpoints and/or the Remote Setpoint. **ATTENTION** “KEY ERROR” will appear in the lower display, if: • the remote setpoint or additional local setpoints are not configured as a setpoint source • you attempt to change the setpoint while a setpoint ramp is enabled, or • if you attempt to change the setpoint with the setpoint select function key disabled. • appears to the left of the active setpoint

4.9 Timer

Introduction

The Timer provides a configurable Time-out period of from 0 to 99 hours:59 minutes or 0 to 99 minutes:99 seconds.

Timer "Start" is selectable as either the **RUN/HOLD** key or Alarm 2.

The Timer display can be either "Time Remaining" or "Elapsed Time".

Configuration check

Make sure:

- TIMER is enabled
- A TIMEOUT period has been selected (in hours and minutes or minutes and seconds)
- A TIMER FUNCTION START has been selected (KEY or AL2)
- A TIMER display has been selected (Time remaining or Elapsed time)
- A timer increment selected
- Timer reset selected

Refer to *Subsection 3.7 Algorithm Set Up Group* for details.

Viewing Times

The times are viewed on the lower display as follows:

TIME REMAINING	will show as a *decreasing* Hrs:Min value (HH:MM) or Min:Sec value (MM:SS) plus a ***counterclockwise*** rotating clock face.
ELAPSED TIME	will show as an *increasing* Hrs:Min value(HH:MM) or Min:Sec value (MM:SS) plus a ***clockwise*** rotating clock face.

Operation

When the Timer is enabled (RUN/HOLD key or ALARM 2), it has exclusive control of Alarm 1 relay.

At "TIME-OUT:

- Alarm 1 is active
- The clock character has stopped moving
- The Time display shows either 00:00 or the time-out period depending on the configuration selection
- The Timer is ready to be reset

At "RESET":

- Alarm 1 relay is inactive
- The time display shows the time-out period
- The time-out period can be changed at this time using the or keys.
- The Timer is ready to be activated

4.10 Accutune III

Introduction

Accutune III (TUNE) may be used for self-regulating and single integrating processes. This autotuning method is initiated on-demand, typically at initial start-up.

There are no other requirements necessary, such as prior knowledge to the process dynamics or initial or post tune process line-out to setpoint or manual output.

Also, the setpoint value is not required to change in order to initiate the tuning process, but the controller must be in the Automatic mode to start tuning. The process need not be in a static (lined out) state and may be dynamic (changing with a steady output).

Configuration check

Make sure:

- TUNE has been enabled see to *Subsection 3.6 - Accutune Set Up Group* for details.

Tuning indicators

A "T" will show in the leftmost alphanumeric of the upper display until tuning is completed.

Operation

The Accutune III algorithm provides user-friendly, on-demand tuning in this controller. No knowledge of the process is required at start-up. The operator simply initiates the tuning while in the automatic mode.

Once Accutune III has been enabled in the TUNE setup group, either “SLOW” or “FAST” tuning may be used. Which one is used is selected via the lower display during normal operation.

For the SLOW selection, the controller calculates conservative tuning constants with the objective of minimizing overshoot. If the controller determines that the process has appreciable dead time, it will automatically default to use Dahlin Tuning, which produces very conservative tuning constants. The SLOW selection may be useful for TPSC and Position Proportional applications, as it reduces “hunt” problems for the motor. This selection is also recommended for applications that have significant deadtimes.

For the FAST selection, the controller calculates aggressive tuning constants with the objective of producing quarter damped response. Depending upon the process, this selection will usually result in some overshoot. For this reason, it may be desireable to enable the FUZZY tune selection. See Section 4.11. When Fuzzy tune is enabled, it will work to suppress or eliminate any overshoot that may occur as a result of the calculated tuning parameters as the PV approaches the setpoint. This selection is best suited for processes with a single lag or for those that do not have any appreciable deadtime. FUZZY tuning does not work well for processes that have appreciable deadtime.

The Accutune III tuning process will cycle the controller’s output two full cycles between the low and high output limits while allowing only a very small Process Variable change above and below the SP during each cycle. A “T” shows in the upper display until tuning is completed.

At the end of the tuning process, the controller immediately calculates the tuning constants and enters them into the Tuning group, and begins PID control with the correct tuning parameters. This works with any process, including integrating type processes, and allows retuning at a fixed setpoint.

4.10.1 Tune for Simplex Outputs

After “TUNE” has been enabled, you can start Accutune as shown in Table 4-10.

Table 4-10 Procedure for Starting “TUNE”

Step	Operation	Press	Result
1	Configure LSP1	Lower Display	Until **SP** (Local Setpoint 1) shows in the lower display.
2		▲ or ▼	Until LSP1 is to the desired value.
3	Switch to “Automatic” Mode	Man Auto	Until the “A” indicator is lighted (on controllers with Manual option).

4	Show Tuning Prompt	Lower Display	Until "TUNE OFF" is shown on lower display.
5	Initiate Tuning	▲	Select "DO SLOW" or "DO FAST" in lower display.
6	Tuning in operation	Lower Display	Upper display will show a "T" as long as ACCUTUNE process is operating. When process completes, tuning parameters are calculated and lower display will show "NO TUNE" prompt.

ATTENTION

The Accutune process may be aborted at any time by changing the lower display back to "NoTUNE" or by switching the controller into Manual Mode.

4.10.2 Tune for Duplex (Heat/Cool)

Accutune for applications using Duplex (Heat/Cool) control.

The controller must be configured to have two local setpoints unless Blended Tuning is desired (see below). See *Subsection 3.11- Control Set Up Group* for details on configuring two local setpoints. During tuning, the Accutune III process assumes that Local Setpoint 1 will cause a Heating demand (output above 50%), and the tuning parameters calculated for that setpoint are automatically entered as PID SET 1. Likewise, Accutune III assumes that Local Setpoint 2 will cause a Cooling demand (output less than 50%), and the tuning parameters calculated for that setpoint are automatically entered as PID SET 2.

Configuration Check for Duplex

See *Subsection 3.6 - Accutune Set Up Group* for details.

Make sure:

- TUNE has been enabled
- DUPLEX has been configured to Manual, Automatic or Disabled

4.10.3 Using AUTOMATIC TUNE at start-up for Duplex (Heat/Cool)

Used when DUPLEX has been configured for AUTOMATIC. This is the preferred selection for most Heat/Cool applications when tuning a new chamber. This selection will sequentially perform both Heat and Cool tuning without further operator intervention.

Table 4-11 Procedure for Using AUTOMATIC TUNE at Start-up for Duplex Control

Step	Operation	Press	Result
1	Configure LSP1	Lower Display	Until **SP** (Local Setpoint 1) shows in the lower display.
2		▲ or ▼	Until LSP1 is a value within the **Heat** Zone (output above 50%).
3	Configure LSP2	Lower Display	Until **2SP** (Local Setpoint 2) shows in the lower display.
4		▲ or ▼	Until LSP2 is a value within the **Cool** Zone (output below 50%).
5	Switch to “Automatic” Mode	Man Auto	Until the “A” indicator is lighted (on controllers with Manual option).
6	Show Tuning Prompt	Lower Display	Until “TUNE OFF” is shown on lower display.
7	Initiate Tuning	▲	Select “DO SLOW” or “DO FAST” in lower display.
	Tuning in operation	Lower Display	Upper display will show a “T” as long as ACCUTUNE process is operating. When process completes, tuning parameters are calculated and lower display will show “NO TUNE” prompt.

4.10.4 Using BLENDED TUNE at start-up for Duplex (Heat/Cool)

When DUPLEX has been configured for DISABLE. This is the preferred selection for Heat/Cool applications which use a highly insulated chamber (a chamber which will lose heat very slowly unless a cooling device is applied). Only one local setpoint (LSP 1) is needed for this selection.

This selection results in performance tuning over the full range utilizing both Heat and Cool outputs to acquire blended tune values that are then applied to both Heat and Cool tuning parameters. Both PID sets are set to the same values.

Table 4-12 Procedure for Using BLENDED TUNE at Start-up for Duplex Control

Step	Operation	Press	Result
1	Configure LSP1	Lower Display	Until **SP** (Local Setpoint 1) shows in the lower display.
2		▲ or ▼	Until the Setpoint is to the desired value.
3	Switch to “Automatic” Mode	Man Auto	Until the “A” indicator is lighted (on controllers with Manual option).
4	Show Tuning Prompt	Lower Display	Until “TUNE OFF” is shown on lower display.
5	Initiate Tuning	▲	Select “DO SLOW” or “DO FAST” in lower display.
6	Tuning in operation	Lower Display	Upper display will show a “T” as long as ACCUTUNE process is operating. When process completes, tuning parameters are calculated and lower display will show “NO TUNE” prompt.

4.10.5 Using MANUAL TUNE at start-up for Duplex (Heat/Cool)

When DUPLEX has been configured for MANUAL. This selection should be used when tuning is needed only for the HEAT zone or only for the COOL zone but not both. If Local Setpoint 1 is used, then the controller will perform a HEAT zone tune. If Local Setpoint 2 is used, then the controller will perform a COOL zone tune.

Table 4-13 Procedure for Using MANUAL TUNE for Heat side of Duplex Control

Step	Operation	Press	Result
1	Configure LSP1	Lower Display	Until **SP** (Local Setpoint 1) shows in the lower display.
2		▲ or ▼	Until LSP1 is a value within the **Heat** Zone (output above 50%).

Step	Operation	Press	Result
3	Switch to “Automatic” Mode	**Man Auto**	Until the “A” indicator is lighted (on controllers with Manual option).
4	Show Tuning Prompt	**Lower Display**	Until “TUNE OFF” is shown on lower display.
5	Initiate Tuning	▲	Select “DO SLOW” or “DO FAST” in lower display.
6	Tuning in operation	**Lower Display**	Upper display will show a “T” as long as ACCUTUNE process is operating. When process completes, tuning parameters are calculated and lower display will show “NO TUNE” prompt.

Table 4-14 Procedure for Using MANUAL TUNE for Cool side of Duplex Control

Step	Operation	Press	Result
1	Configure LSP2	**Lower Display**	Until **2SP** (Local Setpoint 2) shows in the lower display.
2		▲ or ▼	Until LSP2 is a value within the **Cool** Zone (output below 50%).
3	Switch to “Automatic” Mode	**Man Auto**	Until the “A” indicator is lighted (on controllers with Manual option).
4	Show Tuning Prompt	**Lower Display**	Until “TUNE OFF” is shown on lower display.
5	Initiate Tuning	▲	Select “DO SLOW” or “DO FAST” in lower display.
6	Tuning in operation	**Lower Display**	Upper display will show a “T” as long as ACCUTUNE process is operating. When process completes, tuning parameters are calculated and lower display will show “NO TUNE” prompt.

4.10.6 Error Codes

Table 4-15 Procedure for Accessing Accutune Error Codes

Step	Operation	Press	Result
1	Select Accutune Set-up Group	Setup	*Upper Display* = **SETUP** *Lower Display* = **ACCUTUNE**
2	Go to Error Code Prompt	Function	*Upper Display* = **(an error code)** *Lower Display* = **AT ERROR** **Table 4-16 lists all the error codes, definitions, and fixes.**

Table 4-16 Accutune Error Codes

Error Code (Upper Display)	Definition	Fix
RUNNING	ACCUTUNE RUNNING	The Accutune process is still active (Read Only)
NONE	NO ERRORS OCCURRED DURING LAST ACCUTUNE PROCEDURE	None
ID FAIL	PROCESS IDENTIFICATION FAILURE Autotune has aborted because an illegal value of GAIN, RATE, or reset was calculated.	• Illegal Values – try Accutune again. • untunable process -- contact local application engineer.
ABORT	CURRENT ACCUTUNE PROCESS ABORTED caused by the following conditions: ***a.*** *Operator changed to Manual mode* ***b.*** *Digital Input detected* ***c.*** *In Heat region of output and a Cool output calculated or vice versa.*	Try Accutune again
SP2	LSP2 not enabled or LSP1 or LSP2 not in use (only applies to Duplex Tuning)	Enable LSP2 and configure the desired LSP1 and LSP2 setpoints. See *Section 4.10.*

Aborting Accutune

To abort Accutune and return to the last previous operation (SP or output level), press **MAN-AUTO** key to abort the Accutune process or increment from the "DO SLOW" or "DO FAST" prompt to the "TUNE OFF" prompt.

Completing Accutune

When Accutune is complete, the calculated tuning parameters are stored in their proper memory location and can be viewed in the TUNING Set up Group, and the controller will control at the local setpoint using these newly calculated tuning constants.

4.11 Fuzzy Overshoot Suppression

Introduction

Fuzzy Overshoot Suppression minimizes Process Variable overshoot following a setpoint change or a process disturbance. This is especially useful in processes which experience load changes or where even a small overshoot beyond the setpoint may result in damage or lost product.

How it works

The Fuzzy Logic in the controller observes the speed and direction of the PV signal as it approaches the setpoint and temporarily modifies the internal controller response action as necessary to avoid an overshoot. There is no change to the PID algorithm, and the fuzzy logic does not alter the PID tuning parameters. This feature can be independently Enabled or Disabled as required by the application to work with the Accutune algorithm. Fuzzy Tune should not be enabled for processes that have an appreciable amount of deadtime.

Configuration

To configure this item, refer to Section 3 - Configuration:

Set Up Group "**ACCUTUNE**"
Function Prompt "**FUZZY**"
Select "**ENABLE"** or "**DISABLE**" - Use ▲or▼.

4.12 Using Two Sets of Tuning Constants

Introduction

You can use two sets of tuning constants for single output types and choose the way they are to be switched. (this does not apply for Duplex control, which always uses two PID sets).

The sets can be:

- keyboard selected,
- automatically switched when a predetermined process variable value is reached,
- automatically switched when a predetermined setpoint value is reached.

Set up Procedure

The following procedure (Table 4-17) to:

- select two sets,
- set the switch-over value,
- set tuning constant value for each set.

Table 4-17 Set Up Procedure

Step	Operation	Press	Result
1	Select Control Set-up Group	Setup	Until you see: *Upper Display* = **SET** *Lower Display* = **CONTRL**
2	Select PID SETS	Function	Until you see: *Upper Display* = **(available selections)** *Lower Display* = **PID SETS**
3	Select PID SETS Function	▲ or ▼	To select the type of function. Available selections are: **1 ONLY—**1 set of constants **2KEYBD**—2 sets, keyboard selectable **2PV SW**—2 sets, auto switch at PV value **2SP SW**—2 sets, auto switch at SP value
4	Set Tuning Values for Each Set		Refer to "TUNING" Set up group, subsection 3.4 and set the following tuning parameters: PB or GAIN * RATE MIN * RSET MIN or RSET RPM * CYC SEC or CYC SX3 * PB2 or GAIN2 ** RATE2MIN ** RSET2MIN or RSET2RPM ** CYC2SEC or CYC2SX3 ** *PIDSET1 will be used when PV or SP, whichever is selected, is **greater** than the switchover value. **PIDSET2 will be used when PV or SP, whichever is selected, is **less** than the switchover value.
5	Set Switchover Value for **2 PVSW** or **2 SPSW** Selection	Function	Until you see: *Upper Display* = **(the switchover value)** *Lower Display* = **SW VAL**
		▲ or ▼	To select the switchover value in the upper display.

Switch between two sets via keyboard (without automatic switch-over)

Table 4-18 Procedure for Switching PID SETS from the Keyboard

Step	Operation	Press	Result
1	Select Control Set-up Group	Lower Display	Until you see: *Upper Display* = **(the PV value)** *Lower Display* = **PIDS X (X= 1 or 2)**
2		▲ or ▼	To change PID SET 1 to PID SET2 or Vice Versa. You can use Accutune on each set.
3		Lower Display	To accept changes.

4.13 Alarm Setpoints

Introduction

An alarm consists of a relay contact and an operator interface indication. The alarm relay is de-energized if setpoint 1 or setpoint 2 is exceeded.

The alarm relay is energized when the monitored value goes into the allowed region by more than the hysteresis.

The relay contacts can be wired for normally open (NO) energized or normally closed (NC) de-energized using internal jumper placement. See Table 2-3 in the *Section 2 – Installation* for alarm relay contact information.

There are four alarm setpoints, two for each alarm. The type and state (High or Low) is selected during configuration. See *Subsection 3.13 – Configuration* for details.

Alarm Setpoints Display

Table 4-19 Procedure for Displaying Alarm Setpoints

Step	Operation	Press	Result
1	Select Alarm Set-up Group	Setup	Until you see: *Upper Display* = **SET** *Lower Display* = **ALARMS**
2	Access the Alarm Setpoint Values	Function	To successively display the alarm setpoints and their values. Their order of appearance is shown below. *Upper Display* = **(the alarm setpoint value)** *Range values are within the range of the selected parameters except:* ***DEVIATION** (DEV) value = PV Span* ***EVENTS** (EV-ON/EV-OFF) value = Event Segment Number* ***PV RATE OF CHANGE** (PVRATE) = The amount of PV change in one minute in engineering units.* ***LOOP BREAK ALARMS** (BREAK) = The timer value may be changed only for controllers configured for ON/OFF control.* *Lower Display* = **A1S1 VAL =** Alarm 1, Setpoint 1 Value **A1S2 VAL** = Alarm 1, Setpoint 2 Value **A2S1 VAL =** Alarm 2, Setpoint 1 Value **A2S2 VAL** = Alarm 2, Setpoint 2 Value NOTES: *With Three position step control, alarms set for "output" will not function.* *MANUAL, RSP, and F'SAFE selections do not have setpoint values.*
3	Change a value	▲ or ▼	To change any alarm setpoint value in the upper display.
4	Return to Normal Display	Lower Display	

4.14 Three Position Step Control Algorithm

Introduction

The Three Position Step Control algorithm allows the control of a valve (or other actuator) with an electric motor driven by two controller output relays; one to move the motor upscale, the other to move it downscale, without a feedback slidewire linked to the motor shaft.

Estimated Motor Position

The Three Position Step control algorithm provides an output display which is an estimated motor position since there is no slidewire feedback.

- Although this output indication is only accurate to a few percent, it is corrected each time the controller drives the motor to one of its stops (0 % or 100 %).
- It avoids all the control problems associated with the feedback slidewire (wear, dirt, and noise).
- When operating in this algorithm, the output display is shown to the nearest percent (that is, no decimal).

See Motor Travel Time (the time it takes the motor to travel from 0 % to 100 %) in section 3.8.

Motor Position Display

Table 4-20 Procedure for Displaying 3Pstep Motor Position

Step	Operation	Press	Result
1	Access the Displays	Lower Display	Until you see: *Upper Display* = **PV** *Lower Display* = **OT (The estimated motor position in %)**

4.15 Setting a Failsafe Output Value for Restart After a Power Loss

Introduction

If the power to the controller fails and power is reapplied, the controller goes through the power up tests, then goes to a user configured FAILSAFE OUTPUT VALUE.

Set a Failsafe Value

Table 4-21 Procedure for Setting a Failsafe Value

Step	Operation	Press	Result
1	Select Control Set-up Group	**Setup**	Until you see: *Upper Display* = **SET** *Lower Display* = **CONTROL**
2	Select Failsafe Function Prompt	**Function**	You will see: *Upper Display* = **(range)** *within the range of the Output 0 to 100 for all output types except Three Position Step* ***Three Position Step*** *0 = motor goes to closed position* *100 = motor goes to open position* *Lower Display* = **F'SAFE**
3	Select a value	▲ or ▼	To select a failsafe output value in the upper display
4	Return to Normal Display	**Lower Display**	At power up, the output will go to the value set.

4.16 Setting Failsafe Mode

Introduction

You can set the Failsafe Mode to be Latching or Non-Latching.

Set Failsafe Mode

Table 4-22 Procedure for Setting a Failsafe Mode

Step	Operation	Press	Result
1	Select Control Set-up Group	Setup	Until you see: *Upper Display* = **SET** *Lower Display* = **CONTROL**
2	Select Failsafe Function Prompt	Function	You will see: *Upper Display* = **LATCH** (Controller goes to manual and output goes to failsafe value) **NoLATCH** (Controller mode does not change and output goes to failsafe value) *Lower Display* = **FSMODE**
3	Select a value	▲ or ▼	To select a failsafe mode in the upper display.
4	Return to Normal Display	Lower Display	At power up, the output will go to the value set.

4.17 Setpoint Rate/Ramp/Program Overview

Introduction

The Setpoint Ramp configuration group lets you enable and configure any of the following:

- **SP RATE** – a specific rate of change for any local setpoint change. (Subsection 4.18)
- **SP RAMP** – a single setpoint ramp that occurs between the current local setpoint and a final local setpoint over a time interval of 1 to 255 minutes. (Subsection 4.19)
- **SP PROG** – a ramp/soak profile in a 12-segment program. (Subsection 4.20)

This section explains the operation of each selection and configuration reference where necessary.

PV Hot Start

This is a standard feature. At power-up, the setpoint is set to the current PV value and the Rate or Ramp or Program then starts from this value.

RUN/HOLD key

You can start or stop the Ramp or Program using the RUN/HOLD key.

4.18 Setpoint Rate

Introduction

When you have configured a SETPOINT RATE, it will apply immediately to local setpoint change.

Configuration check

Make sure:

- SPRATE is enabled
- A Rate Up (EUHRUP) or Rate Down (EUHRDN) value has been configured in Engineering units per hour.

ATTENTION
A value of 0 will imply an immediate change in setpoint, that is, NO RATE applies. *See Subsection 3.5 – Configuration group "SPRAMP" for details.)*

Operation

When a change to local setpoint is made, this controller will ramp from the original setpoint to the "target" setpoint at the rate specified.

The current setpoint value is shown as **SPn XXXX** on the lower display while the "target" setpoint is shown as **SP XXXX** on the lower display.

Power outages

If power is lost before the "target" setpoint is reached, upon power recovery, the controller powers up with Sn = Current PV value and it automatically "Restarts" from Sn = current PV value up to the original "target" setpoint.

4.19 Setpoint Ramp

Introduction

When you have configured a SETPOINT RAMP, the ramp will occur between the current local setpoint and a final local setpoint over a time interval of from 1 to 255 minutes. You can RUN or HOLD the ramp at any time.

Configuration Check

Make sure

- SPRAMP is enabled
- SP RATE and SPPROG are not running.

- A Ramp Time (TIMIN) in minutes has been configured
- A final setpoint value (FINLSP) has been configured. See Subsection 3.5 – Configuration group "SPRAMP" for details.

Operation

Running a Setpoint Ramp includes starting, holding, viewing the ramp, ending the ramp and disabling it. See Table 4-23.

Table 4-23 Running A Setpoint Ramp

Step	Operation	Press	Result
1	Select Automatic Mode	Man Auto	"A" indicator is on. *Upper Display* = "H" and PV value *Lower Display* = SP and Present value
2	Set Start Setpoint	Lower Display	Until start SP value is in lower display *Upper Display* = "H" and PV value *Lower Display* = SP and start SP value
3	Start the Ramp	Run Hold	You will see *Upper Display* = "R" and a changing PV value *Lower Display* = SP and a changing SP value increasing or decreasing toward the final SP value
4	Hold/Run the Ramp	Run Hold	This holds the ramp at the current setpoint value. Press again to continue.
5	View the remaining ramp time	Lower Display	Until you see *Upper Display* = PV value *Lower Display* = **RAMPXXXM** (time remaining in minutes)
6	End the Ramp		When the final setpoint is reached, "R" changes to "H" in the upper display and the controller operates at the new final setpoint.
7	Disable SPRAMP		See Section 3 – Configuration group "SPRAMP" for details.

Power Outage

If power is lost during a ramp, upon power-up the controller will be in HOLD and the setpoint value will be the setpoint value prior to the beginning of the setpoint ramp.

The ramp is placed in hold at the beginning.

Configure the mode at Set Up Group "CONTROL", function prompt "PWR MODE". See Subsection 3.11 – CONTROL SETUP GROUP Prompts.

4.20 Setpoint Ramp/Soak Programming

Introduction

The term "programming" is used here to identify the process for selecting and entering the individual ramp and soak segment data needed to generate the required setpoint versus time profile (also called a program).

A segment is a ramp or soak function which together make up a setpoint program. Setpoint Ramp/Soak Programming lets you configure six ramp and six soak segments to be stored for use as one program or several small programs. You designate the beginning and end segments to determine where the program is to start and stop.

Review program data and configuration

While the procedure for programming is straightforward, and aided by prompts, we suggest you read "Program Contents". Table 4-24 lists the program contents and an explanation of each to aid you in configuration. Then refer to Subsection 3.5–Configuration to do the setpoint program.

Make sure SPRAMP is disabled.

Fill out the worksheet

Refer to the example in Figure 4-3 and draw a Ramp/Soak Profile on the worksheet provided (Figure 4-4) and fill in the information for each segment. This will give you a record of how the program was developed.

Operation

Refer to Table 4-25 Run/Monitor the program.

Program Contents

Table 4-24 lists all the program contents and a description of each.

Table 4-24 Program Contents

Associated Prompts	Contents	Definition
STRT SEG	**Start segment number**	The start segment number designates the number of the first segment. Range = 1 to 11
END SEG	**End segment number**	The end segment number designates the number of the last segment, it must be a soak segment (even number). Range = 2 to 12
RECYCLES	**Recycle number**	The recycle number allows the program to recycle a specified number of times from beginning to end. Range = 0 to 99
STATE	**Program state**	The program state selection determines the program state after completion. The selections are: • **DISABLE** = program is disabled (so program value changed to DISABLE) • **HOLD** = program on hold
PROG END	**Program termination state**	The program termination state function determines the status of the controller upon completion of the program. The selections are: • **LAST** = controls to last setpoint • **FAILSAFE** = manual mode and failsafe output.
KEYRESET (ToBEGIN)	**Reset Program to Beginning**	When enabled, this selection allows you to reset the program via the keyboard to the beginning of the program.
KEYRESET (RERUN)	**Rerun current segment**	When enabled, this selection allows you to reset the program via the keyboard to the beginning of the current segment.
HOTSTART	**Hot Start**	This function determines whether LSP1 or PV is used as the setpoint when the program is initially changed from HOLD to RUN. The selections are: **DISABLE** = When the program is initially changed from HOLD to RUN the present LSP1 value is captured as the default setpoint. If the program is terminated or the power cycled before the program has completed, the LSP1 is used as the control setpoint. The beginning segment uses this value as the initial ramp setpoint. **ENABLE** = When the program is initially changed from HOLD to RUN the present PV value is captured and used as the beginning setpoint value for the ramp segment. If the program is terminated before completion, the setpoint value will revert back to the PV value captured at the initial HOLD to RUN transition. If the power is cycled before program completion, upon power-up the setpoint is set to

Associated Prompts	Contents	Definition
		the PV value at power-up and when the program is restarted that setpoint value is used initially.
RAMPUNIT **SEGxRAMP or SEGxRATE**	**Ramp time or rate segments**	A ramp segment is the time it will take to change the setpoint to the next setpoint value in the program. Ramps are odd number segments (1, 3, . . . 11). Segment #1 will be the initial ramp time. Ramp time is determined in either: TIME* - Hours.Minutes Range = 0-99hr.59 min. or RATE* - EU/MIN or EU/HR Range = 0 to 999 * This selection of time or rate is made at prompt “RAMPUNIT”. Set this prompt before entering any Ramp values. **ATTENTION** Entering “0” implies an immediate step change in setpoint to the next soak.
SEGx SP SEGxTIME	**Soak segments**	A soak segment is a combination of soak setpoint (value) and a soak duration (time). • Soaks are even number segments (2, 4, . . . 12). • Segment 2 will be the initial soak value and soak time. • The soak setpoint range value must be within the setpoint high and low range limits in engineering units. • Soak time is the duration of the soak and is determined in: TIME - Hours:Minutes Range = 0-99 hr:59 min.
SOAK DEV	**Guaranteed soak**	All soak segments can have a deviation value of from 0 to ± 99 (specified by SOK DEV) which guarantees the deviation value for that segment. **Guaranteed** soak deviation values greater than zero guarantee that the soak segment’s process variable is within the ± deviation value for the configured soak time. Whenever the ± deviation value is exceeded, the soak timer stops until the process variable gets within the ± deviation value. There are no guaranteed soaks whenever the deviation value is configured to 0, (that is, soak segments start timing soak duration as soon as the soak setpoint is first reached, regardless of where the process variable remains relative to the soak segment). The soak deviation value is the number in engineering units, above or below the setpoint, outside of which the timer halts. The range is 0 to ± 99.XX. The decimal location here corresponds decimal configuration chosen in the Display Set up group.

Ramp/soak profile example

Before you perform the actual configuration, we recommend that you draw a Ramp/Soak profile in the space provided on the "*Program Record Sheet*" (Figure 4-4) and fill in the associated information. An example of a Ramp-Soak Profile is shown in Figure 4-3. Start setpoint is at 200 degrees F.

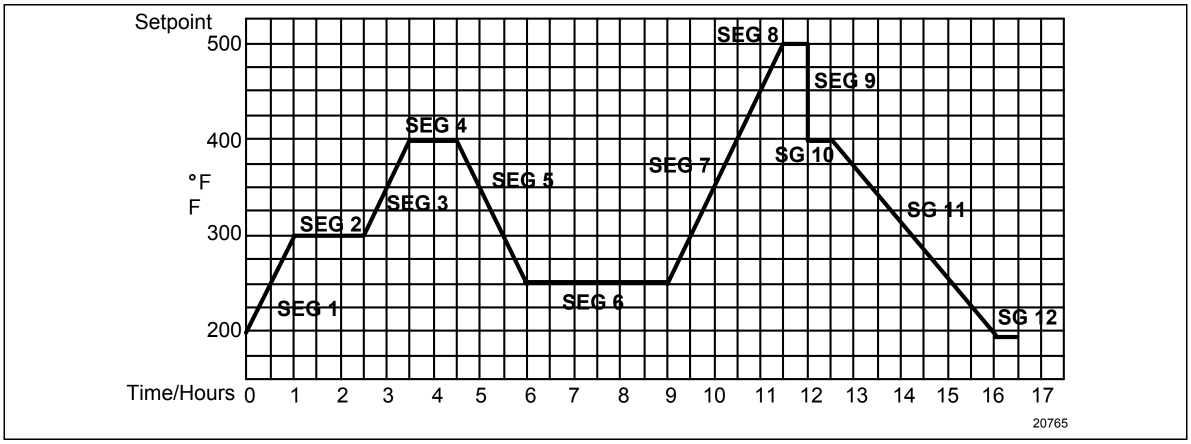


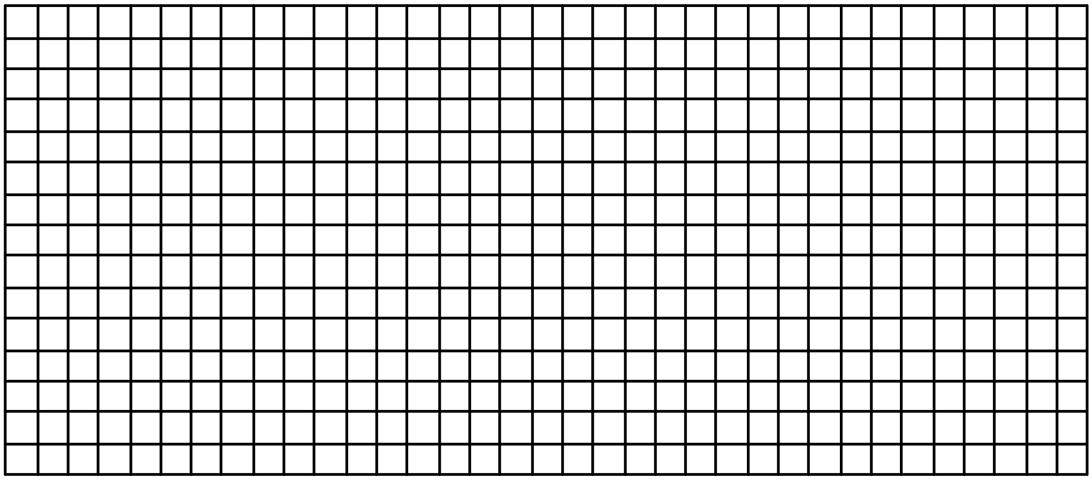
Figure 4-3 Ramp/Soak Profile Example

Ramp/Soak Profile Example

Prompt	Function	Segment	Value
STRT SEG	Start Seg.		1
END SEG	End Seg.		12
RAMP UNIT	Engr. Unit for Ramp		TIME
RECYCLES	Number of Recycles		2
SOAK DEV	Deviation Value		0
PROG END	Controller Status		LAST SP
STATE	Controller State at end		HOLD
KEYRESET	Reset SP Program		DISABLE
HOTSTART	PV Hot Start Program Initialization or power up in SPP		DISABLE
SEG1RAMP	Ramp Time	1	1 hr.
SEG2 SP	Soak SP	2	300
SEG2TIME	Soak Time	2	1 hr:30 min.
SEG3RAMP	Ramp Time	3	1 hr.
SEG4 SP	Soak SP	4	400
SEG4TIME	Soak Time	4	1 hr.
SEG5RAMP	Ramp Time	5	1 hr:30 min.
SEG6 SP	Soak SP	6	250
SEG6TIME	Soak Time	6	3 hr:0 min.
SEG7RAMP	Ramp Time	7	2 hr:30 min.
SEG8 SP	Soak SP	8	500
SEG8TIME	Soak Time	8	0 hr:30 min.
SEG9RAMP	Ramp Time	9	0
SG10 SP	Soak SP	10	400
SG10 TIME	Soak Time	10	0 hr:30 min.
SG11RAMP	Ramp Time	11	3 hr:30 min.
SG12 SP	Soak SP	12	200
SG12TIME	Soak Time	12	0 hr:30 min.

Program record sheet

Draw your ramp/soak profile on the record sheet shown in Figure 4-4 and fill in the associated information in the blocks provided. This will give you a permanent record of your program and will assist you when entering the Setpoint data.



20766

Figure 4-4 Program Record Sheet

Prompt	Function	Segment	Value
STRT SEG	Start Seg.		
END SEG	End Seg.		
RAMPUNIT	Engr. Unit for Ramp		
RECYCLES	Number of Recycles		
SOAK DEV	Deviation Value		
PROG END	Controller Status		
STATE	Controller State at end		
KEYRESET	Reset SP Program		
HOT START	PV Hot Start Program Initialization or power up in SPP		
SEG1RAMP	Ramp Time	1	
SEG2 SP	Soak SP	2	
SEG2TIME	Soak Time	2	
SEG3RAMP	Ramp Time	3	
SEG4 SP	Soak SP	4	

Prompt	Function	Segment	Value
SEG4TIME	Soak Time	4	
SEG5RAMP	Ramp Time	5	
SEG6 SP	Soak SP	6	
SEG6TIME	Soak Time	6	
SEG7RAMP	Ramp Time	7	
SEG8 SP	Soak SP	8	
SEG8TIME	Soak Time	8	
SEG9RAMP	Ramp Time	9	
SG10 SP	Soak SP	10	
SG10 TIME	Soak Time	10	
SG11RAMP	Ramp Time	11	
SG12 SP	Soak SP	12	
SG12TIME	Soak Time	12	

Run/Monitor the program

Prior to running the program, make sure all the “SP PROG” function prompts under the Set Up group “SP RAMP” have been configured with the required data.

“H” appears in the upper display indicating that the program is in the HOLD state.

ATTENTION SP Program parameter *cannot* be changed during RUN state; the unit must be in the HOLD state in order to change parameters.

Run/Monitor functions

Table 4-25 lists all the functions required to run and monitor the program.

Table 4-25 Run/Monitor Functions

Function	Press	Result
Set the Local Setpoint	Lower Display	*Upper Display* = PV value *Lower Display* = **SP**
	▲ or ▼	To set the Local Setpoint value to where you want the program to start out.
Run State	Run Hold	Initiates the setpoint program. “**R**” appears in the upper display indicating that the program is running.
Hold State	Run Hold	Holds the setpoint program. “**H**” appears in the upper display indicating that the program is in the HOLD state. The setpoint holds at the current setpoint.
External Hold		If one of the Digital Inputs is programmed for the HOLD function, then contact closure places the controller in the HOLD state, if the setpoint program is running. The upper display will periodically show “**H**” while the switch is closed. ATTENTION The keyboard takes priority over the external switch for the RUN/HOLD function. Reopening the HOLD switch runs the program.
Viewing the present ramp or soak segment number and time	Lower Display until you see	*Upper Display* = PV value *Lower Display* = **XXRAHH.MM** for Ramps or = **XXSKHH.MM** for Soaks Time remaining in the SEGMENT in hours and minutes. XX = The segment number, 1 to 12.

continued

Function	Press	Result
Viewing the number of cycles left in the program	Lower Display until you see	*Upper Display* = PV value *Lower Display* = **RECYC XX** Number of cycles remaining in the setpoint program. X = 0 to 99
End Program		When the final segment is completed, the "**R**" in the upper display either changes to "**H**" (if configured for HOLD state), or disappears (if configured for disable of setpoint programming). • The controller then either operates at the last setpoint in the program or goes into manual mode/failsafe output, depending upon the "LAST" configuration.
Disable Program		See Section 3 – Configuration Group "SP PROG" for details.

Power outage

ATTENTION If power is lost during a program, upon power-up the controller will be in hold and the setpoint value will be the setpoint value prior to the beginning of the setpoint program. The program is placed in hold at the beginning. The mode will be as configured under "PWR UP" in the "CONTROL" group.

Digital input (remote switch) operation

Program can be placed in RUN, HOLD, RERUN, or BEGIN state through a remote dry contact connected to optional digital input terminals, as follows:

RUN—contact closure places Program in RUN state, OR

HOLD—contact closure places Program in HOLD state

RERUN—contact closure allows the Setpoint Programmer to be reset to the initial segment of its current cycle, unit stays in previous mode.

Opening the contact will cause the Controller to revert to its original state.

BEGIN— Contact closure resets SP Program back to the beginning of the first segment in the program and places the program in the HOLD mode. Program cycle number is not affected. Reopening switch has no effect.

Opening the contact will cause the Controller to revert to its original state.